WORKED SOLUTION

STEP I 2002, Question 5

Pure • Integer Roots of Polynomials

> **STEP I 2002, Question 5**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/02-s1-q5

The key idea

A polynomial written as a product of linear factors is easy to evaluate at special points: putting in $x = 0$, $x = 1$ or $x = -1$ turns the factorised form into a product of the k_i, $k_i + 1$ or $k_i - 1$, while the expanded form gives a sum of coefficients. For the quartic, those values are fixed integers, and integer roots mean every $k_i \pm 1$ must be a divisor of a known number. Factorising those numbers leaves very little room.

Solution

The three identities

Since the two expressions for $f(x)$ are the same polynomial, they agree at every value of x.

Putting $x = 0$: the expanded form gives $f(0) = a_n$, since every other term contains a factor of x. The factorised form gives $f(0) = k_1 k_2 \cdots k_n$. Hence

$$k_1 k_2 \cdots k_n = a_n.$$

Putting $x = 1$: every power of x becomes 1, so the expanded form gives $f(1) = 1 + a_1 + a_2 + \cdots + a_n$, while the factorised form gives $(1 + k_1)(1 + k_2) \cdots (1 + k_n)$. Hence

$$(k_1 + 1)(k_2 + 1) \cdots (k_n + 1) = 1 + a_1 + a_2 + \cdots + a_n,$$

as required.

Putting $x = -1$: the factorised form gives $(k_1 - 1)(k_2 - 1) \cdots (k_n - 1)$. In the expanded form, the term $a_j x^{n-j}$ becomes $(-1)^{n-j} a_j = (-1)^n (-1)^j a_j$ (using $(-1)^{-j} = (-1)^j$), so

$$(k_1 - 1)(k_2 - 1) \cdots (k_n - 1) = (-1)^n (1 - a_1 + a_2 - a_3 + \cdots + (-1)^n a_n).$$

For even n this is simply $1 - a_1 + a_2 - \cdots + a_n$; for odd n every sign is reversed.

The quartic

Write the quartic as $f(x) = (x+k_1)(x+k_2)(x+k_3)(x+k_4)$, so its roots are $-k_1, \ldots, -k_4$, and these are integers. Using the identities with $n = 4$:

$$k_1k_2k_3k_4 = 576 = 2^6 \cdot 3^2,$$
$$(k_1+1)(k_2+1)(k_3+1)(k_4+1) = 1 + 22 + 172 + 552 + 576 = 1323 = 3^3 \cdot 7^2,$$
$$(k_1-1)(k_2-1)(k_3-1)(k_4-1) = 1 - 22 + 172 - 552 + 576 = 175 = 5^2 \cdot 7.$$

Every k_i is at least 2. All the coefficients are positive, so $f(x) > 0$ for every $x \geq 0$; hence every root is negative and every k_i is a positive integer. Moreover no k_i equals 1, since otherwise the third product would be 0, not 175. So $k_i \geq 2$ for each i, and the numbers $k_i - 1$ and $k_i + 1$ are all positive integers.

? **Ask yourself** How do we turn three products into a short list of possible k_i?

Each $k_i - 1$ is a positive divisor of 175, and each $k_i + 1$ is a positive divisor of 1323. The divisors of 175 are $1, 5, 7, 25, 35, 175$, giving

$$k_i \in \{2, 6, 8, 26, 36, 176\}, \qquad k_i + 1 \in \{3, 7, 9, 27, 37, 177\}.$$

Of these values of $k_i + 1$, only $3, 7, 9, 27$ divide $1323 = 3^3 \cdot 7^2$ (since 37 is prime and $177 = 3 \cdot 59$). So each k_i is one of $2, 6, 8, 26$, and each $k_i + 1$ is one of $3, 7, 9, 27$.

Now look at the product $\prod(k_i + 1) = 3^3 \cdot 7^2$.

- The factor 7^2 can only come from factors equal to 7, so exactly two of the $k_i + 1$ equal 7: two of the k_i are 6.
- The other two factors are powers of 3 from $\{3, 9, 27\}$ with product $3^3 = 27$. Each is at least 3, so 27 is impossible (the partner would be 1), and the only option is 3×9. So the remaining k_i are 2 and 8.

Thus $\{k_1, k_2, k_3, k_4\} = \{2, 6, 6, 8\}$. As a check against the first product, $2 \cdot 6 \cdot 6 \cdot 8 = 576$, as required. Expanding confirms it:

$$(x+2)(x+8) = x^2 + 10x + 16, \qquad (x+6)^2 = x^2 + 12x + 36,$$

and

$$(x^2 + 10x + 16)(x^2 + 12x + 36) = x^4 + 22x^3 + 172x^2 + 552x + 576.$$

So the roots are

$$x = -2, \quad x = -6 \text{ (a repeated root)}, \quad x = -8.$$

Checks

- The sum of the k_i is $2 + 6 + 6 + 8 = 22$, matching the coefficient of x^3.

- $f(-6) = 1296 - 4752 + 6192 - 3312 + 576 = 0$, and $f'(x) = 4x^3 + 66x^2 + 344x + 552$ also vanishes at $x = -6$, confirming that -6 is a double root.
- A quick numerical test of the identities: for $f(x) = (x+2)(x-3) = x^2 - x - 6$ we get $(2-1)(-3-1) = -4$ and $1 - (-1) + (-6) = -4$.

Common slips

- Getting the sign wrong in the $x = -1$ result: the factor $(-1)^n$ matters when n is odd.
- Forgetting that a repeated root is allowed: once 7^2 is seen in 1323, two of the k_i must be equal.
- Listing only positive divisors without first showing that every $k_i \geq 2$; that step is what makes all the $k_i \pm 1$ positive.

Beyond STEP

The identities here are special cases of Vieta's formulae, which express the coefficients of a polynomial as the elementary symmetric functions of its roots. The divisor argument is a cousin of the rational root theorem, and of Gauss's lemma on factorising integer polynomials, which you meet in a first course on rings. Allenby's *Rings, Fields and Groups* develops these ideas carefully.

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